

TDD using JUnit5 and Mockito

Module: JUnit_Basic Testing Exercises | Exercise 1: Setting Up JUnit

Scenario

You need to set up JUnit in your Java project to start writing unit tests.

Steps Followed

1. Created a new Java project using Maven's standard directory layout.
2. Added the JUnit 5 (JUnit Jupiter) dependency to the project's pom.xml, along with Mockito for future mocking-based tests.
3. Created a test class (CalculatorTest) containing unit tests for a sample Calculator class.

Project Structure

```
JUnitBasicTestingExercises/  
|-- pom.xml  
|-- src/  
    |-- main/java/com/example/Calculator.java  
    |-- test/java/com/example/CalculatorTest.java
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"  
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
    http://maven.apache.org/xsd/maven-4.0.0.xsd">  
  <modelVersion>4.0.0</modelVersion>  
  
  <groupId>com.example</groupId>  
  <artifactId>JUnitBasicTestingExercises</artifactId>  
  <version>1.0-SNAPSHOT</version>  
  <packaging>jar</packaging>  
  
  <properties>  
    <maven.compiler.source>21</maven.compiler.source>  
    <maven.compiler.target>21</maven.compiler.target>  
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>  
  </properties>  
  
  <dependencies>  
    <dependency>  
      <groupId>org.junit.jupiter</groupId>  
      <artifactId>junit-jupiter</artifactId>  
      <version>5.10.2</version>  
      <scope>test</scope>  
    </dependency>  
    <dependency>  
      <groupId>org.mockito</groupId>  
      <artifactId>mockito-junit-jupiter</artifactId>  
      <version>5.11.0</version>  
      <scope>test</scope>  
    </dependency>  
  </dependencies>  
</project>
```

```

        </dependency>
</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-surefire-plugin</artifactId>
            <version>3.2.5</version>
        </plugin>
    </plugins>
</build>
</project>

```

Calculator.java

```

package com.example;

public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int divide(int a, int b) {
        if (b == 0) {
            throw new ArithmeticException("Cannot divide by zero");
        }
        return a / b;
    }
}

```

CalculatorTest.java

```

package com.example;

import org.junit.jupiter.api.Test;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.DisplayName;
import static org.junit.jupiter.api.Assertions.*;

class CalculatorTest {

    private Calculator calculator;

    @BeforeEach
    void setUp() {
        calculator = new Calculator();
    }

    @Test
    @DisplayName("Adding two positive numbers")
    void testAdd() {
        assertEquals(5, calculator.add(2, 3));
    }

    @Test

```

```
@DisplayName("Subtracting two numbers")
void testSubtract() {
    assertEquals(1, calculator.subtract(3, 2));
}

@Test
@DisplayName("Dividing by zero throws exception")
void testDivideByZero() {
    assertThrows(ArithmeticException.class, () -> calculator.divide(10, 0));
}
}
```

How to Run

Terminal: `mvn test`

VS Code: open `CalculatorTest.java` and click the 'Run Test' link above the class name, or use the Testing panel.

Expected Result

```
Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
BUILD SUCCESS
```